

AI Palmistry & Tarot Intelligence Platform

Milestone 2 - Palm Analysis & Tarot Intelligence

1. Palm Analysis Engine

- The final approach adopted an open-source deep learning model (yeonsumia/palmistry) that uses a trained U-Net to detect real Heart, Head, and Life lines, after MediaPipe view-invariant image alignment. Three files in the repository were patched to work with current MediaPipe/OpenCV versions.

This final approach was tested on 300 real images from the dataset, achieving a measured 91.3% success rate (274/300).

2. Tarot Interpretation Engine

- A 78-card tarot dataset (22 Major Arcana, 56 Minor Arcana) was used to build a card-drawing function that selects cards without repetition, assigns upright/reversed orientation, and generates interpretation text from the dataset's own light/shadow meanings.

3. Reading Generation & Reports

- A combined function merges one hand's palm reading with a 3-card tarot spread into a single structured report, exported as a PDF.

Milestone 3 — AI Interpretation & Recommendations

1. LLM Integration

- Groq's API (llama-3.3-70b-versatile model) was integrated to synthesize palm and tarot data into natural, personalized text, replacing the fixed-template sentences used in Milestone 2. The API key is stored securely using Colab's Secrets manager.

2. AI Interpretation Engine

- Given a hand's palm line data and a drawn tarot spread, the engine generates a single coherent narrative reading (150-200 words) that weaves both sources together, plus separate Past, Present, and Future descriptions tied specifically to each tarot card in the spread.

3. Personality Intelligence Module

- Alongside the narrative, the model returns structured data: personality traits, strengths, and growth areas (deliberately framed constructively), extracted directly from the same palm and tarot facts.

4. Recommendation Engine

- A second AI call turns the personality profile into four categorized, actionable recommendations: personal growth, relationship guidance, career suggestion, and goal alignment.

5. Life Trend Analysis Engine

- A third AI call analyzes the full profile and tarot spread to produce a life path theme, a current opportunity, a constructively-framed potential challenge, and a growth potential statement.

Tools & Libraries Used

- MediaPipe Tasks API, PyTorch (U-Net), OpenCV, scikit-image — palm detection
- Groq API (llama-3.3-70b-versatile) — AI interpretation, recommendations, life trend analysis
- Python, Pandas — data handling
- fpdf2, Pillow — combined PDF report generation with embedded images
- Google Colab (GPU) + Google Drive — development and storage

Summary

- In Milestone 2, we built a palm analysis engine using a deep-learning model (91.3% accuracy on real images) and a 78-card tarot engine, combining both into downloadable reading reports.
- In Milestone 3, we integrated Groq's LLM to synthesize palm and tarot data into personalized narratives, structured personality insights, recommendations, and life trend analysis. Finally, we combined both milestones' outputs into a single PDF report showing the user's name, palm readings, tarot cards with images, and past/present/future predictions.

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